

DAY 15 — CTF LEARNING REPORT

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1. Which challenge was easiest, and why?

The easiest challenge for me was the basic Linux file-navigation and file-reading task. Commands such as `pwd`, `ls`, `cd` and `cat` are straightforward once their purpose is understood. I could move through directories, list files and inspect their contents without needing a complicated technique. This challenge felt easier because it tested foundational skills that I had already practised.

2. Which challenge was hardest, and what made it hard?

The most difficult part for me was interpreting the clues and deciding what to investigate next, especially when access or file permissions were involved. At first, I was not always sure whether the issue was the file location, the current user, or the permissions. I learned that guessing commands is less useful than checking the evidence step by step—for example, confirming the current directory, examining file details and then choosing an appropriate next action.

3. Which Linux commands did you use during the CTF?

I used `pwd` to show my current directory; `ls` and `ls -la` to list normal and hidden files; `cd` to change directories; `cat` to read text files; `find` to locate files; `grep` to search text; `file` to identify file types; `whoami` and `id` to check the current user and groups; and `chmod` to understand or change permissions when appropriate. I also used DNS utilities such as `nslookup` where DNS information was relevant, and `Nmap` to examine network ports and services in the authorised lab. Each command helped answer a specific question rather than being run randomly.

4. What did you learn about permissions?

Linux permissions control which users can read, write or execute a file, and which users can access a directory. The three permission classes are owner, group and others. The symbols r, w and x represent read, write and execute. I learned to inspect permissions with `ls -l` and to interpret numeric modes such as 755: the owner has read, write and execute permissions, while group and others have read and execute permissions. I also learned that directory permissions behave differently from file permissions, so permission problems should be checked carefully.

5. What did you learn about DNS?

DNS, or Domain Name System, translates human-readable domain names into IP addresses and helps clients locate services. I learned that different DNS record types serve different purposes: A maps a name to an IPv4 address, AAAA maps it to IPv6, MX identifies mail servers, CNAME provides an alias, NS identifies name servers, and TXT stores text information. Tools such as `nslookup` can help query DNS information. DNS knowledge is useful during reconnaissance because it can reveal how a domain's services are configured.

6. What did you learn about ports?

A port is a numbered endpoint used by network applications and services to communicate. I learned that common services often use well-known ports, for example SSH on 22, DNS on 53, HTTP on 80, HTTPS on 443 and MySQL on 3306. An open port generally means a service is accepting connections; closed means no service is listening, while filtered means a network device or rule prevents Nmap from determining the state. A port number alone does not prove that a system is vulnerable—the service and its configuration also matter.

7. What did you learn from reading Nmap output?

I learned to read Nmap output as evidence about a target rather than just a list of numbers. The host-status information indicates whether a host appears reachable. The PORT, STATE and SERVICE columns show the port number and protocol, its observed state, and the service Nmap associates with it. When version detection is used in an authorised lab, Nmap may also report product or version details. I learned that results depend on scan type, network conditions and filtering, so they should be interpreted carefully and verified where necessary.

8. What would you revise before the next CTF?

Before the next CTF, I would revise Linux navigation and file commands, hidden-file discovery, grep and find, file types, ownership and permissions, DNS record types, common TCP/UDP ports, and the meaning of Nmap states and service output. I would also practise reading clues carefully, keeping notes of commands and results, and changing one variable at a time when troubleshooting. Most importantly, I would practise in authorised beginner CTF labs so that I can build confidence without relying on guesswork.

Conclusion

This CTF helped me connect cybersecurity theory with practical investigation. I improved my understanding of Linux commands, permissions, DNS, ports and Nmap output. The main lesson for me was to work methodically: understand the clue, gather information, interpret the output and then decide on the next step. I will continue practising these fundamentals in authorised labs before moving to more advanced challenges.